

Update 1.04

HERON GNSS-Reflectometry Graduate Project

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Austin Hunter, Andras Szalay**

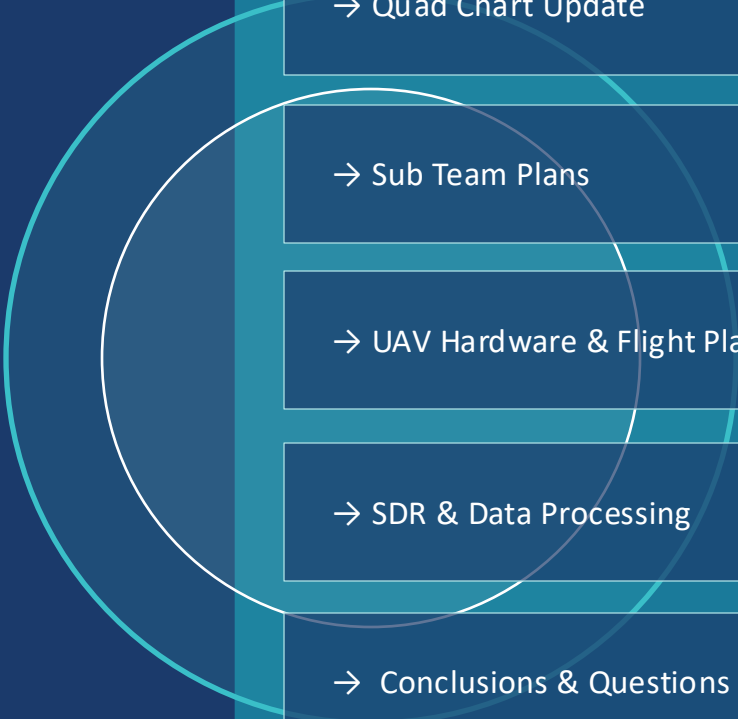
Smead Aerospace Engineering Sciences Dept. · CCAR · University of Colorado Boulder

September 9, 2026



1.04

Update Outline



→ Quad Chart Update

→ Sub Team Plans

→ UAV Hardware & Flight Planning

→ SDR & Data Processing

→ Conclusions & Questions

Week Update: Sept. 9

UAV Hardware &
Flight Planning

SDR & Data
Processing

Accomplished This Week

Planned for the Next Week

✓ Insert here

✓ Insert here

Current Challenges

❑ Insert here

❑ Insert here

Proposed Solutions

✓ TRADE STUDIES &
EXTERNAL EXPERTISE

UAV Hardware & Flight Planning

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Software Defined Radio (SDR) & Data Processing

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UAV Software

UAV Mission Planner

Specular Point Mission Planner:

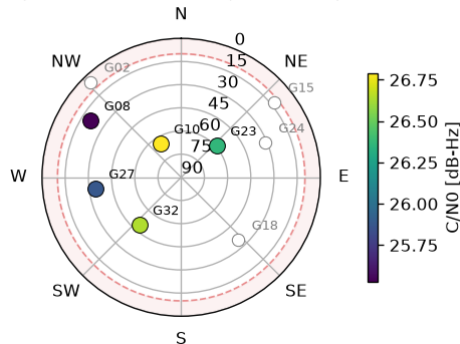
UAV Flight Readiness

Ground Test Reflections – potential site selection

New Test with B200 at lower bandwidth & # of bits

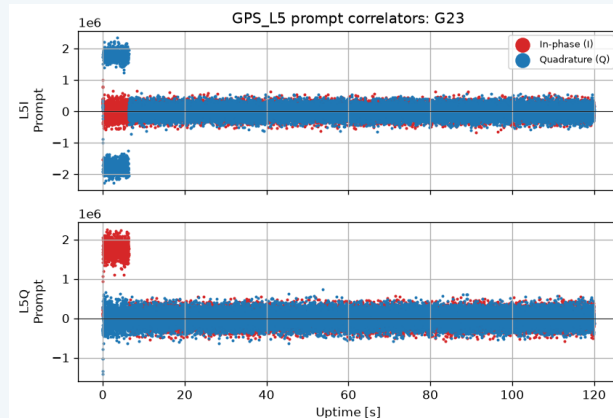
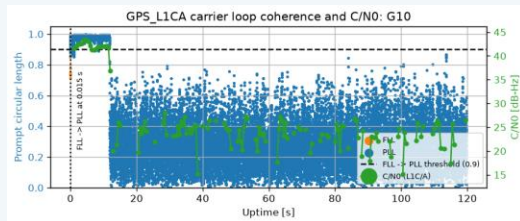
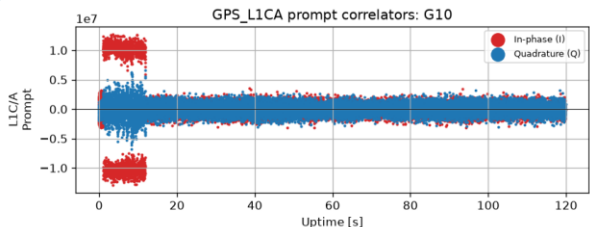
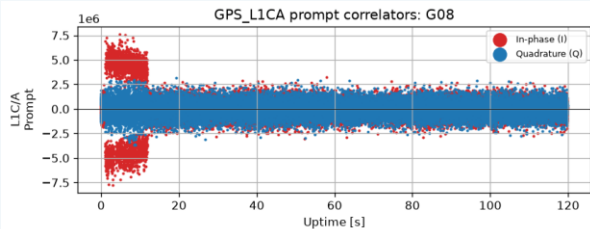
Live Sky Test 09.14

20260914_150800_l1_rhcp_120s
(9 above the horizon, 5 tracked)



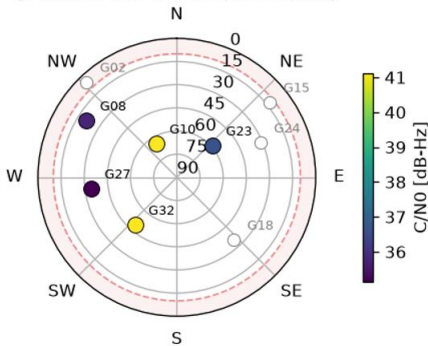
SDR was encountering overflow errors during logging, code does not reacquire or track through these outages.

```
Press Ctrl + C to stop streaming...
OGot an overflow indication. Please consider the following:
Your write medium must sustain a rate of 100.000000MB/s.
Dropped samples will not be written to the file.
Please modify this example for your purposes.
This message will not appear again.
0000000000
```

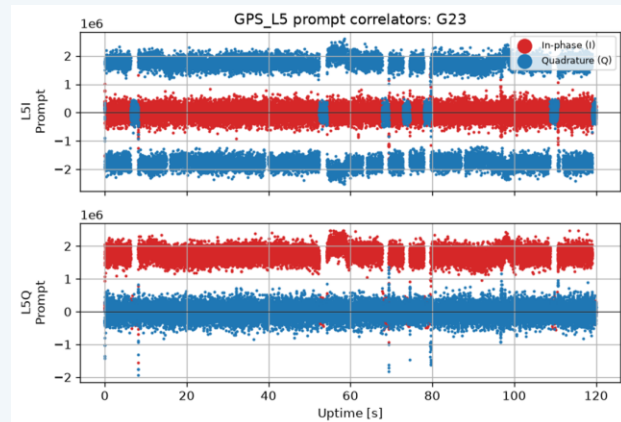
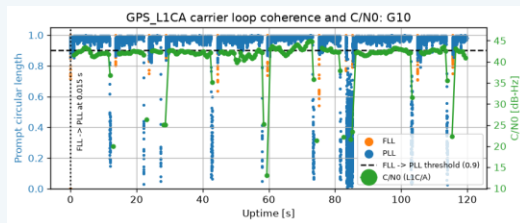
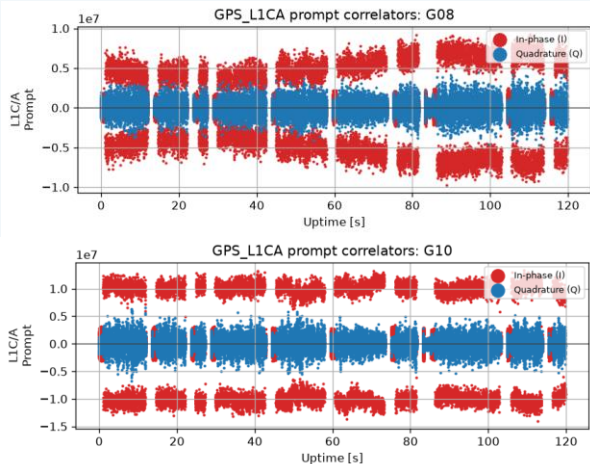


Live Sky Test 09.14

20260914_150800_l1_rhcp_120s
(9 above the horizon, 5 tracked)

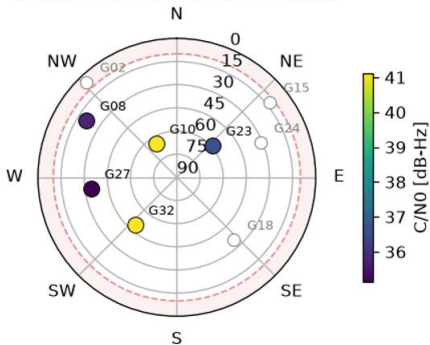


Not continuous enough for nav data bits on L5, (need a longer tracking period without dropped packets/an improved algorithm)

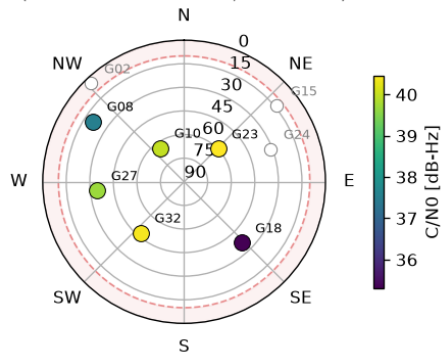


Live Sky Test 09.14

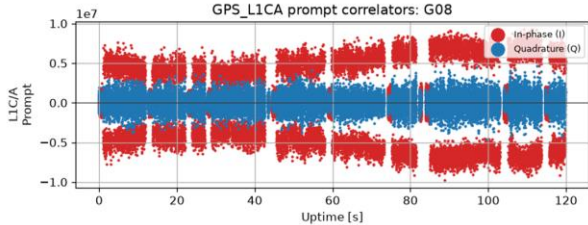
20260914_150800_L1_rhcp_120s
(9 above the horizon, 5 tracked)



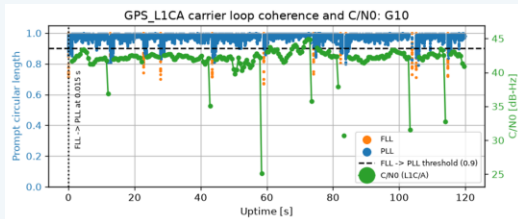
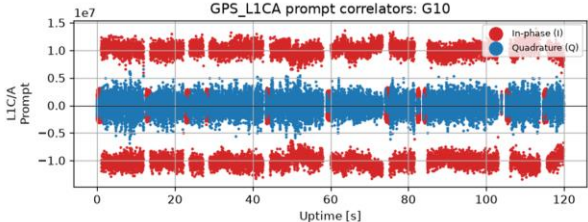
20260914_150800_L5_rhcp_120s
(9 above the horizon, 6 tracked)



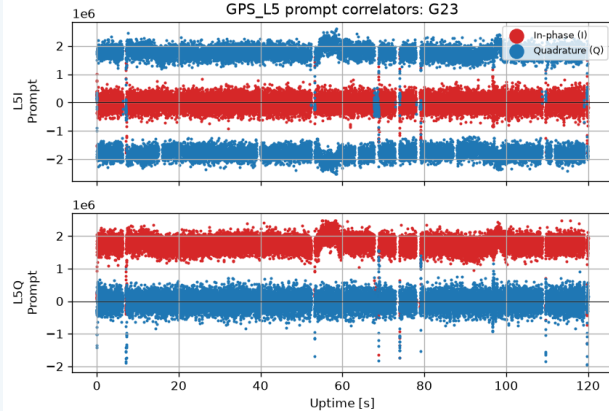
GPS_L1CA prompt correlators: G08



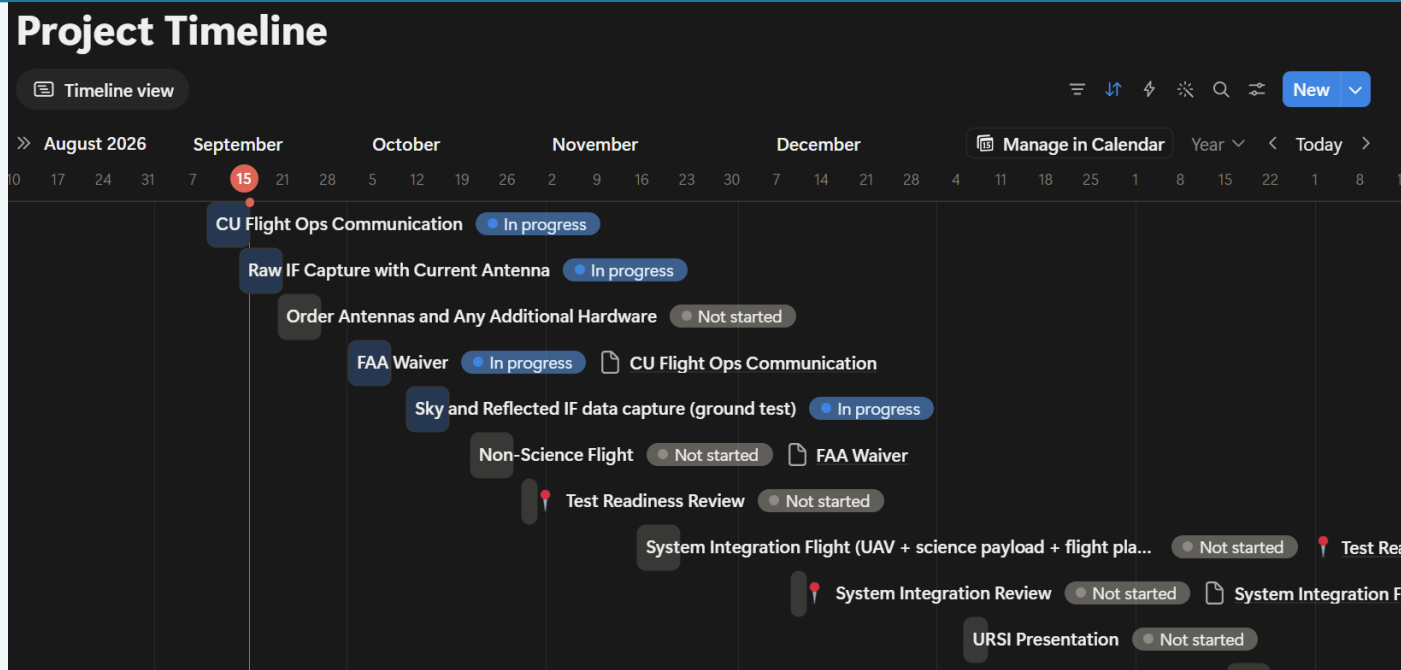
GPS_L1CA prompt correlators: G10



GPS_L5 prompt correlators: G23



Updated Schedule with TRR + SIR



Proposed Review Dates for

Grad Projects

**Test Readiness Review
(TRR) Wed. 10/28**

**System Integration Review
(SIR) Wed. 12/09**

Plan for the next week

Overall Objectives

- Finish URSI Abstract Submission (internal deadline Thursday for draft)
- Add finalized semester review Presentations to calendar
- Determine if we have CAD drawing file from SURGE & update system block diagram

UAV Hardware & Flight Planning

- Complete battery capacity/health testing
- Renew FAA registration of UAV
- Finalize Weight & Balance and submit updated documentation to CU Flight Ops
- Finalize preparations for the initial non-science UAV flight / system checkout
- Complete custom Mission Planning Software

&

Software Defined Radio (SDR) & Data Processing

- Place Order(s) for SDR and clock
- Perform full sky data collect
- Improve SDR logging code to make more user friendly for data processing
- Begin antenna comparison/characterization

Questions?

Questions & Discussion

